

Problema 1:

Inciso a)

```
In[1]:= Edo1 = y''[x] + y[x] == Sin[2 x];
Condiciones1 = {y[0] == 2, y'[0] == 1};
Sol1 = DSolve[{Edo1, Condiciones1}, y[x], x];
Solgeneral1 = y[x] /. Sol1[[1]];
Simplify[Solgeneral1]
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$$\text{Out[5]} = \frac{1}{3} (6 \cos[x] + 5 \sin[x] - 2 \cos[x] \sin[x])$$

Inciso b)

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In[11]:= Edo2 = y'''[x] - 4 y''[x] - 7 y'[x] + 10 y[x] == x * Exp[2 x];
Condiciones2 = {y[0] == 0, y'[0] == 0, y''[0] == 0};
Sol2 = DSolve[{Edo2, Condiciones2}, y[x], x];
Solgeneral2 = y[x] /. Sol2[[1]];
Simplify[Solgeneral2]
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$$\text{Out[14]} = \frac{e^{-2x} (3 - 84 e^{3x} + 77 e^{4x} + 4 e^{7x} - 84 e^{4x} x)}{1008}$$

$$\text{Out[15]} = \frac{e^{-2x} (3 - 84 e^{3x} + 4 e^{7x} + e^{4x} (77 - 84 x))}{1008}$$

Problema 3:

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In[24]:= Sistema1 = {x''[t] - 4 x[t] - 2 y'[t] + y[t] == t, 2 x'[t] + x[t] + y''[t] == 0};
Sol3 = DSolve[Sistema1, {x[t], y[t]}, t];
Solgeneralx = x[t] /. Sol3[[1]];
Solgeneraly = y[t] /. Sol3[[1]];
Simplify[Solgeneralx]
Simplify[Solgeneraly]
```

$$\text{Out[28]} = \frac{1}{4} e^{-t} (7 c_1 + 3 e^{2t} c_1 - c_2 + e^{2t} c_2 - c_3 - e^{2t} c_3 + 3 c_4 + e^{2t} c_4 - 2 e^t (3 c_1 - c_3 + 2 c_4) \cos[t] + 2 e^t (2 c_1 + c_2 + c_4) \sin[t])$$

$$\text{Out[29]} = \frac{1}{4} e^{-t} (8 e^t + 4 e^t t + 7 c_1 - 9 e^{2t} c_1 - c_2 - 3 e^{2t} c_2 - c_3 + 3 e^{2t} c_3 + 3 c_4 - 3 e^{2t} c_4 + 2 e^t (c_1 + 2 c_2 + c_3) \cos[t] + 2 e^t (8 c_1 + c_2 - 2 c_3 + 5 c_4) \sin[t])$$